

Content Safety Classification for Generative AI Systems

Politecnico di Torino

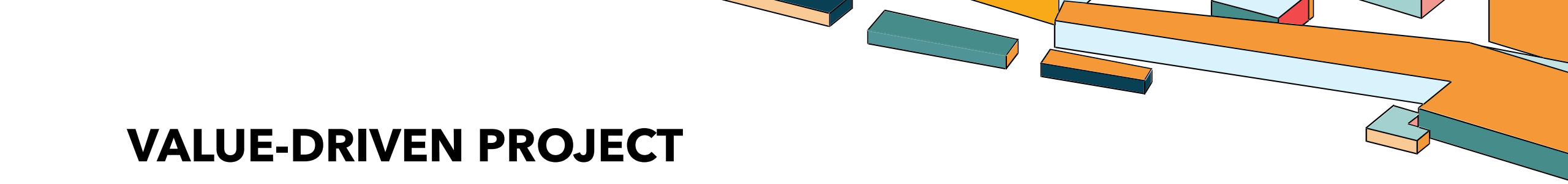
FONDAZIONE links

PASSION FOR INNOVATION

PROJECT PRESENTATION

This project focuses on developing a content safety classifier: a model that reads the prompt sent to a generative system and the response it returns, and labels them as safe or unsafe together with the harm categories violated.





VALUE-DRIVEN PROJECT

Any system that generates text for the public needs a gate between the model and the user. This project builds and benchmarks that gate: an open, measurable classifier that stops harmful output without blocking legitimate content.

Responsible deployment

A safety decision that can be inspected, logged and appealed.

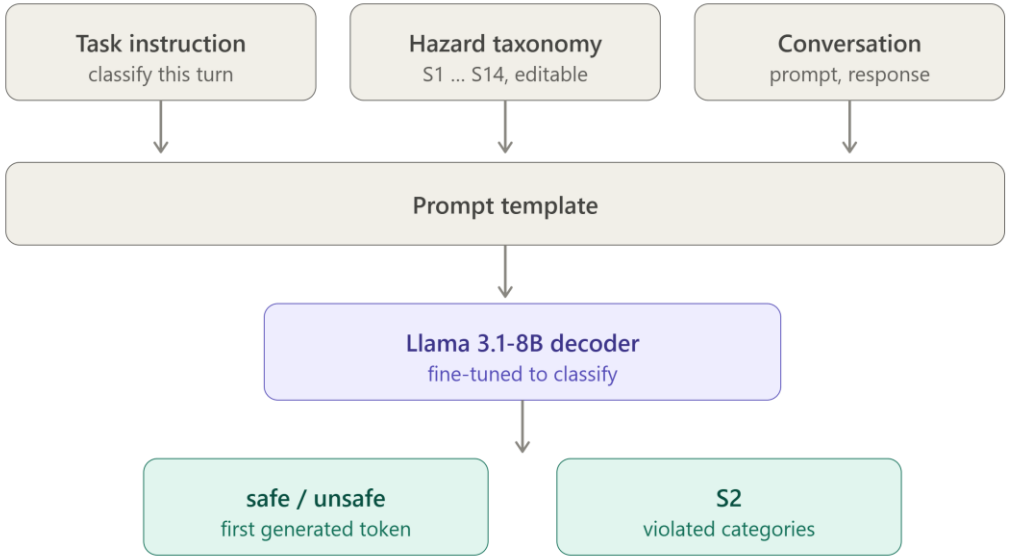
Regulatory readiness

Risk and transparency duties are moving from guidance to obligation.

METHOD

The reference method: an instruction-tuned LLM used as a classifier.

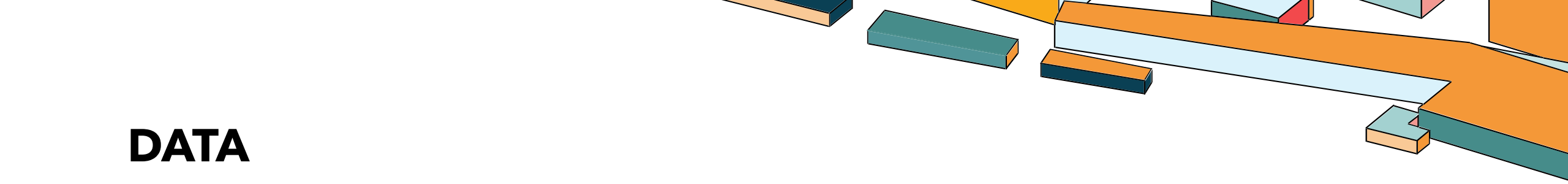
- LLM Model: Llama Guard 3-8B, 14 categories. It generates “safe” or “unsafe” plus the codes.
- The taxonomy is written into the prompt, not into the weights
- **LoRA adaptation** specialises it to the target policy



BASELINE TO COMPARE

Encoder classifier

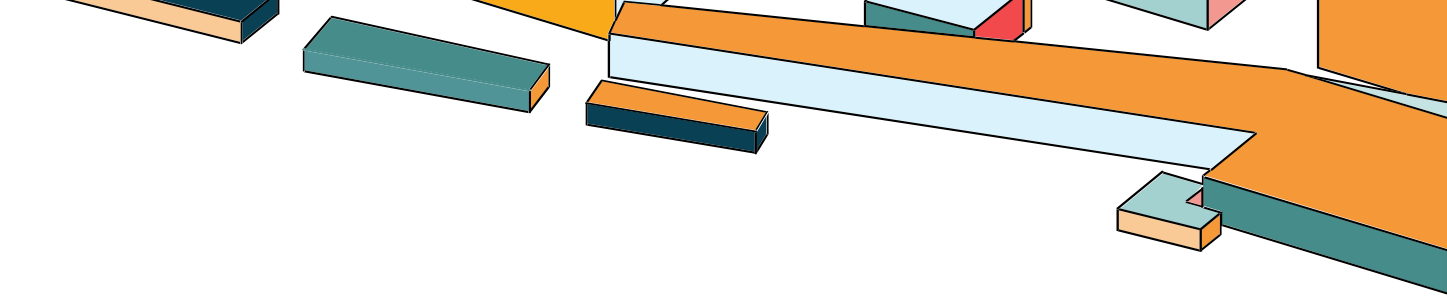
DeBERTa fine-tuned per category



DATA

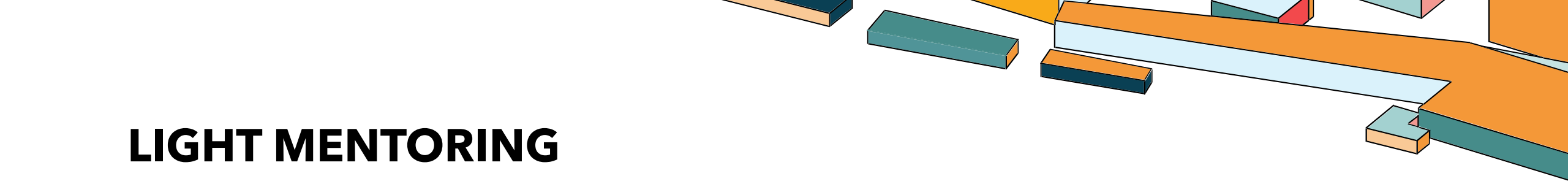
Public datasets of human-labelled prompts and model responses.

- **BeaverTails**
  ~300k QA pairs, labelled over 14 harm categories
- **WildGuardMix**
  Prompt harm, response harm and refusal labels



OBJECTIVES

- Align the taxonomies of the public safety datasets
- Evaluate the guardrail baselines zero-shot
- Fine-tune a guard model with PEFT on the target policy
- Benchmark accuracy, false positives and over-refusal

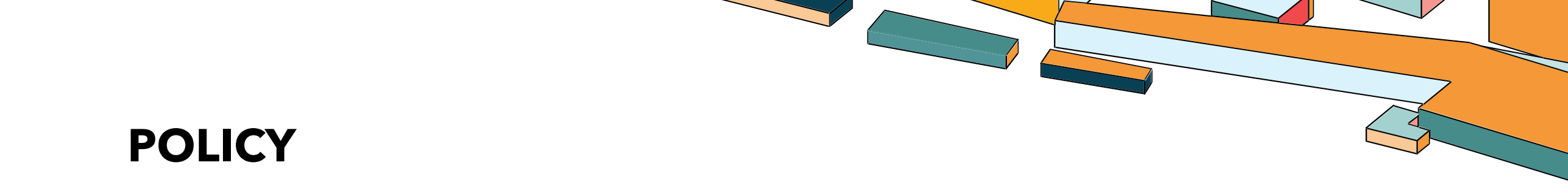


LIGHT MENTORING

- **First (longer) meeting** as an introduction to the existing pipeline and methodology, context and data presentation.

- **Weekly one-hour calls** with students for the whole duration of the semester.

- Feel free to reach out via MS Teams or email at any time for any questions or doubts.



POLICY

- Both project descriptions and implementations will be part of a repository group published on GitHub.

- The repositories will be public unless requests from the organization that will be discussed.

CONTACTS

Federico D’Asaro
federico.dasaro@linksfoundation.com
Juan José Márquez Villacís
juan.marquez@linksfoundation.com

